

Lily Oforiwa Sam

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EDUCATION

Columbia University in the City of New York
Computer Science

New York, NY
August 2019 - May 2023

Relevant Coursework: Intro to Java, Intro to Python for Engineers, Data Structures and Algorithms, Discrete Mathematics, Advanced Programming, Computational Linear Algebra, Databases, User Interface Design, Advanced Software Engineering

HONORS & AWARDS

- **John W. Kluge Scholarship** – awarded to top 5% of Columbia College Admitted students class.
- **Columbia College Dean's List** - awarded per semester for 3.6 or better GPA in 12 or more points of letter credit.
- **Code2040 Finalist (2021/2022)** - Chosen from 1,000+ applicants for highly competitive technical accelerator for Black and Latinx technologists.
- **Twitter DevelopHER 2022 Finalist** - 1 of 30 students selected from 1,200+ applicants; won the #ReadySetCode team programming challenge.

WORK EXPERIENCE

Deutsche Bank Global Technology
Software Engineer

Cary, NC
July 2023 – July 2026

Career Highlights

Instant Payments Tribe Promotion

March 2026

- **Promoted** from Analyst (Software Engineer I) to Associate (**Software Engineer II**) in recognition of consistent delivery of robust payment services and demonstrated success in mentoring junior engineers.

Instant Payments AI Innovation Day Hackathon

March 2026

- Developed AI Agent to automate OSS remediations by writing GitHub Actions workflow to use GitHub Copilot AI tool to access JFrog Xray scans of packaged code and remediate vulnerabilities.

Instant Payments Tribe Contributions

- Building multiple Java SpringBoot microservices to support new instant payments scheme for US and Brazil.
- Saved 20,000+ engineering hours annually by building robust CI/CD pipelines with automated workflows to check controls that enabled on-demand same-day production releases and eliminated manual release overhead.
- Integrating real time payment services with MongoDB Enterprise and Oracle ExaCC Database.
- Setting up monitoring and observability for services and code components on Google Cloud Platform and New Relic.
- Lead on building CI/CD pipelines to support GitHub-based deployment workflows across development, UAT, SIT, and PROD environments.
- Writing Terraform scripts to provision Kubernetes resources on Google Kubernetes Engine (GKE) and GCP.
- Writing OracleSQL stored procedures and functions to build payment processing channels for storing and auditing transactions.
- Built ETL pipelines to process static account and customer data for instant payments, and designed data replication architecture to isolate time-critical payment flow operations from non-time-critical operations.
- Evaluating new surveillance and fraud marker detection model technology to ensure accuracy of models.

Columbia University in the City of New York
Teaching Assistant – COMS 4111

New York, NY
Sep 2022 – May 2023

- Designed course in Python programming language to cover relational and non-relational databases and instructed over 450 students in a graduate-level databases course along with grading exams and assignments.
- Held weekly office hours in Python, Flask and taught databases in MySQL, MongoDB and Neo4j to assist students with understanding course concepts.

Columbia University Center for Student Advising
Tutor

New York, NY
Sep 2021 – May 2023

- Held online private tutoring sessions for students to help them better understand data structures in Python and Java.
- Assisted students with mastering their course content and maximizing their potential for academic achievement.

Amazon Inc.

Software Development Engineer Intern (Full Stack)

Seattle, WA

Jun 2022 – Aug 2022

- Developed payments service API enabling business enterprises and customers to access health insurance contract data.
- Built frontend portal with React and TypeScript using Amazon's Harmony framework and AWS Polaris library.
- Deployed and maintained payments web service on AWS infrastructure including ECS, Fargate, S3, and DynamoDB.

Sponsors of Educational Opportunity (SEO)

Tech Developer Intern

Remote

Jun 2021 – Aug 2021

- Built a full stack web app to query data from the New York Times API using Python and Flask.
- Implemented development, testing, and automation tools to support product infrastructure.

PROJECTS

Text-to-SQL LLM Calibration Evaluator

Python, SQLite, Pandas, Matplotlib, Groq API

- Engineered an automated evaluation framework to measure the Expected Calibration Error (ECE) and confidence reliability of open-source LLMs on the academic Spider Text-to-SQL benchmark.
- Designed an execution-match testing pipeline using Python and local SQLite databases to execute zero-shot AI-generated queries, bypassing the limitations of standard string-comparison grading.
- Quantified AI overconfidence in Llama-3.1-8B (ECE of 0.2840, 98.4% avg. confidence vs. 70.0% execution accuracy) and generated reliability diagrams visualizing the confidence-accuracy gap.
- Currently extending the framework to compare calibration behavior across multiple model families via the Groq API.

ConverterPro (Open-Source Python Package)

Python, Poetry, GitHub Actions, PyPI, LLM APIs

- Architected and published a scalable library for deterministic unit and currency conversions to PyPI, managing the complete deployment lifecycle and API design.
- Implemented robust CI/CD pipelines utilizing GitHub Actions, Poetry for dependency management, and Black for automated code formatting to ensure high reliability.
- Designed a clean, developer-friendly API surface with comprehensive unit testing to guarantee the accuracy of algorithmic data transformations across multiple measurement systems.
- Currently expanding the package to include an LLM-powered Natural Language Processing (NLP) wrapper to enable dynamic conversions via unstructured text inputs.

TECHNICAL SKILLS

Languages & Frameworks: Python, Java, JavaScript, TypeScript, HTML, CSS, React, AngularJS, Flask, Spring Boot, Maven, Apache Kafka, Bash

Cloud & DevOps: Google Cloud Platform (GKE, Cloud Storage, Secret Manager), AWS (ECS, Fargate, S3, DynamoDB), Terraform, Docker, Kubernetes, GitHub, New Relic, GitHub Actions

Databases: MySQL, MongoDB, Neo4j, Oracle SQL

Tools & Methodologies: JIRA, Git, Agile/Scrum, BDD (Gherkin, Cucumber)